

Spin-Dependent Exotic Interactions

Matter–Antimatter Asymmetry Analysis Report

Generated: 2026-05-14 08:02:11

Metric	Value
Pairs analysed	7
Pairs skipped	0
Analysis timestamp	2026-05-14 08:02:11
Framework	SPINDEP v1.0
Method	log-interpolated chi-squared asymmetry

Summary of All Pairs

Coupling	Potential	Sector	Matter source	Antimatter source	A mean	chi2	p-value
gsgs	UNKNOWN	ee	Delaunay2017	Adkins2022	0.873	93298.8	0.000e+00 **
gAgA	V1	ep	Karshenboim2011	Fadeev2022	1.000	119956.2	0.000e+00 **
gAgA	V2	ee	Kotler2015	Fadeev2022	1.000	119989.9	0.000e+00 **
gAgA	V2	ee	Ficek2017	Fadeev2022	0.999	119758.9	0.000e+00 **
gAgA	V2	ee	Ficek2017	Fadeev2022	0.954	109194.9	0.000e+00 **
gAgA	V2	ee	Rong2018	Fadeev2022	1.000	119992.9	0.000e+00 **
gAgA	V2	ee	Jiao2019	Fadeev2022	1.000	119989.9	0.000e+00 **

Pair 1/7 | gsgs · UNKNOWN · ee

Matter dataset	Delaunay2017	Antimatter dataset	Adkins2022
Interaction class	lepton-lepton	Lambda range	8.17e-13 – 1.88e-09 m

Statistic	Value	Interpretation
Mean A_alpha	0.8727	Strong asymmetry — large CPT-sensitive difference.
chi-squared	93298.831	dof = 300
p-value	0.0000e+00	*** (p < 0.001)
lambda min	8.169e-13 m	
lambda max	1.879e-09 m	

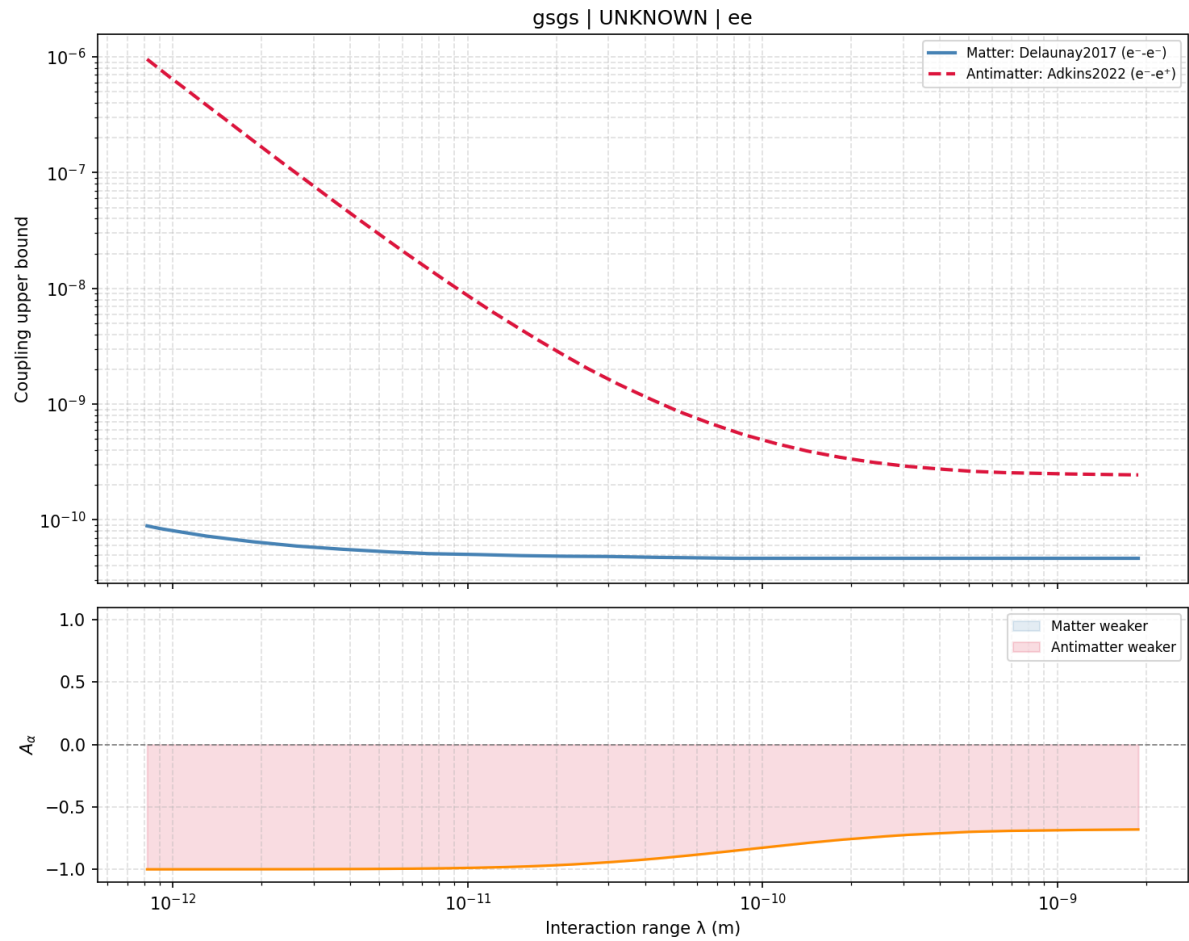


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under UNKNOWN potential.

Pair 2/7 | gAgA · V1 · ep

Matter dataset	Karshenboim2011	Antimatter dataset	Fadeev2022
Interaction class	lepton-nucleon	Lambda range	2.44e-11 – 1.93e-08 m

Statistic	Value	Interpretation
Mean A_alpha	0.9998	Strong asymmetry — large CPT-sensitive difference.
chi-squared	119956.230	dof = 300
p-value	0.0000e+00	*** (p < 0.001)
lambda min	2.439e-11 m	
lambda max	1.929e-08 m	

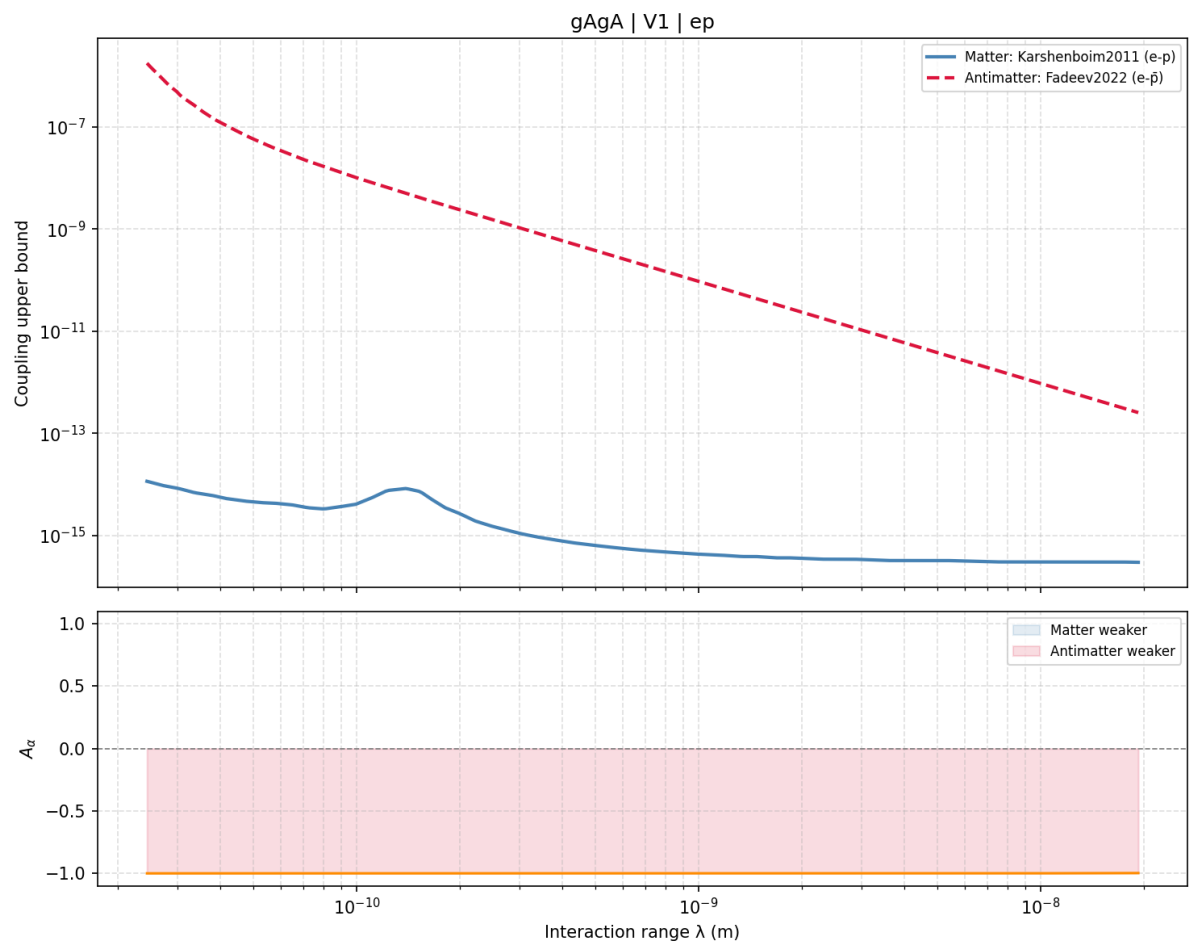


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ep sector under V1 potential.

Pair 3/7 | gAgA · V2 · ee

Matter dataset	Kotler2015	Antimatter dataset	Fadeev2022
Interaction class	lepton-lepton	Lambda range	1.75e-07 – 1.63e-05 m

Statistic	Value	Interpretation
Mean A_alpha	1.0000	Strong asymmetry — large CPT-sensitive difference.
chi-squared	119989.912	dof = 300
p-value	0.0000e+00	*** (p < 0.001)
lambda min	1.754e-07 m	
lambda max	1.629e-05 m	

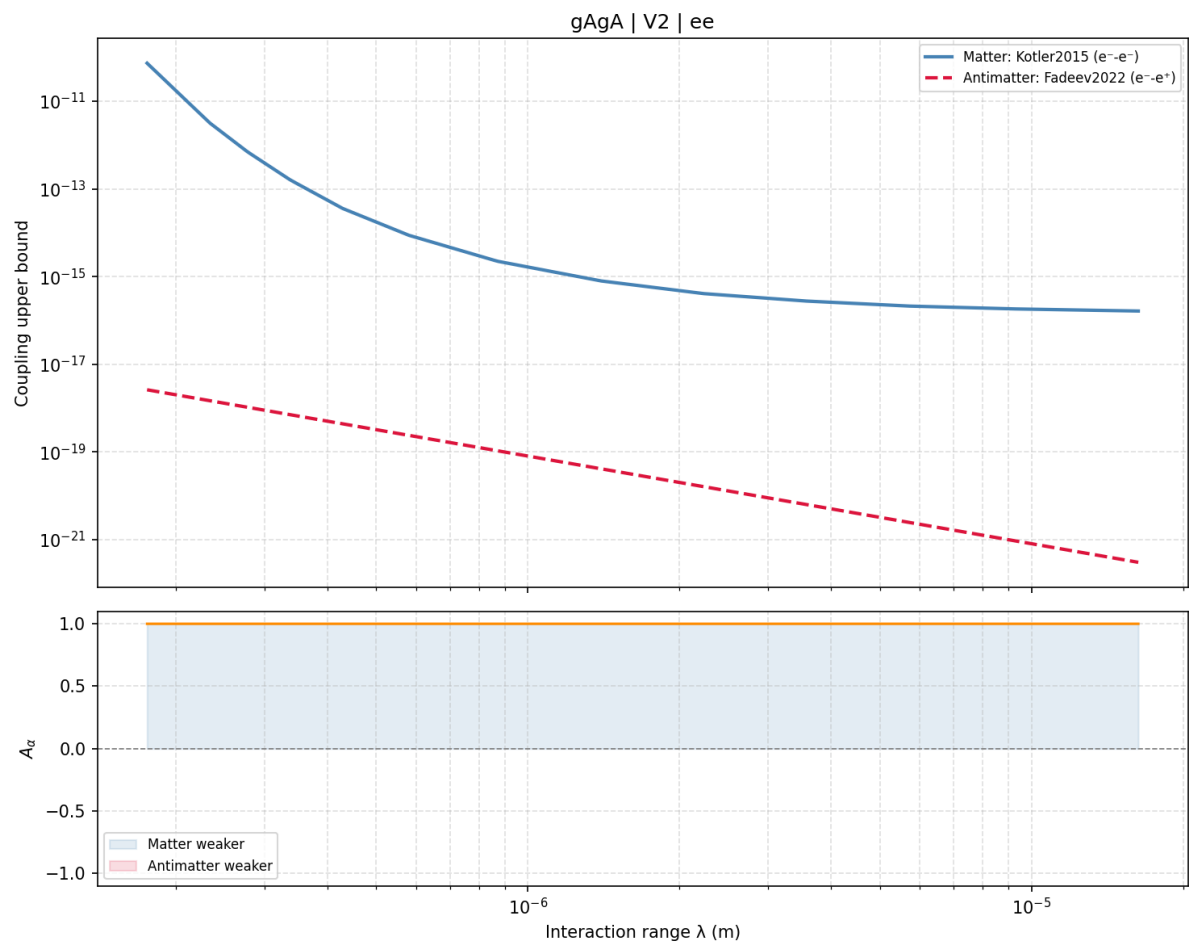


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under V2 potential.

Pair 4/7 | gAgA · V2 · ee

Matter dataset	Ficek2017	Antimatter dataset	Fadeev2022
Interaction class	lepton-lepton	Lambda range	1.97e-11 – 1.31e-05 m

Statistic	Value	Interpretation
Mean A_alpha	0.9990	Strong asymmetry — large CPT-sensitive difference.
chi-squared	119758.872	dof = 300
p-value	0.0000e+00	*** (p < 0.001)
lambda min	1.970e-11 m	
lambda max	1.310e-05 m	

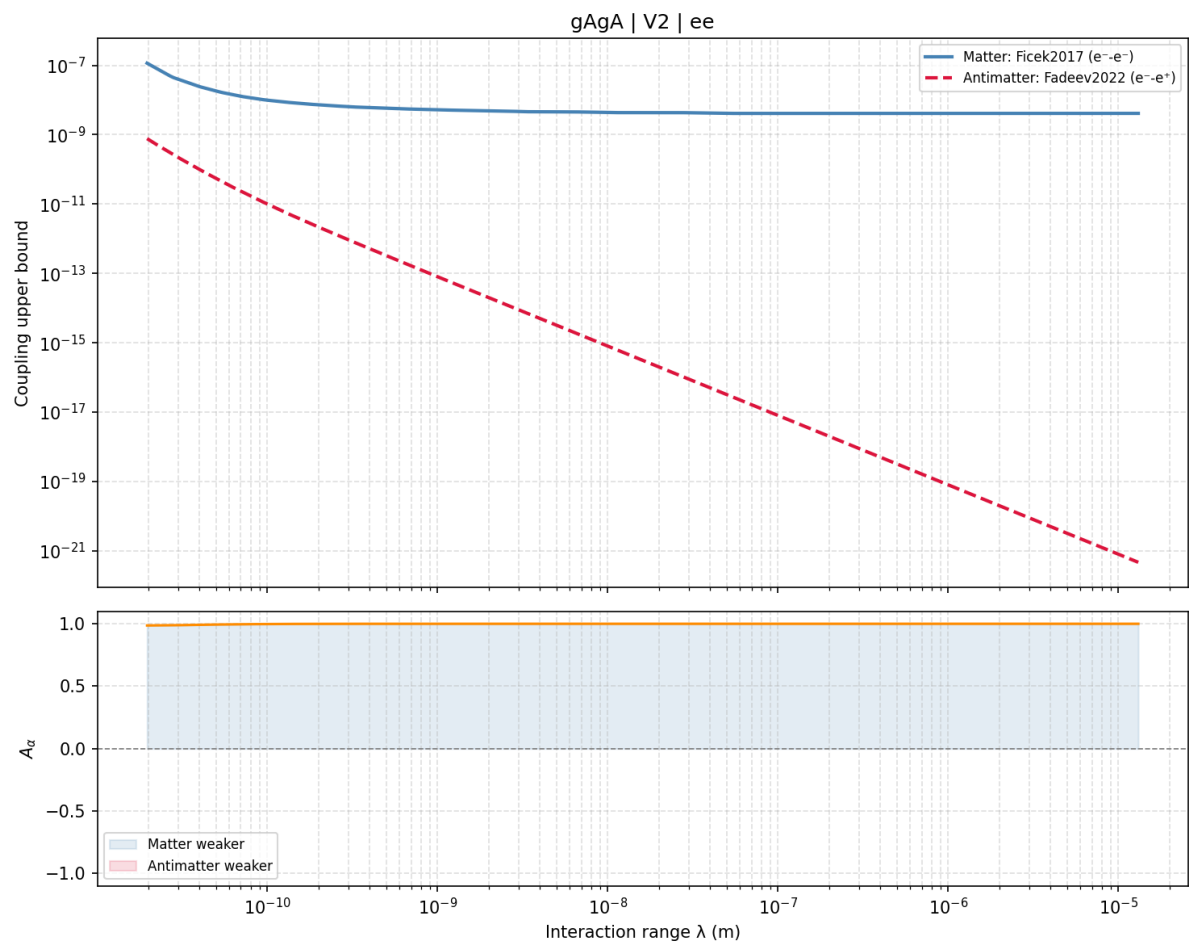


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under V2 potential.

Pair 5/7 | gAgA · V2 · ee

Matter dataset	Ficek2017	Antimatter dataset	Fadeev2022
Interaction class	lepton-lepton	Lambda range	2.50e-12 – 1.63e-05 m

Statistic	Value	Interpretation
Mean A_alpha	0.9539	Strong asymmetry — large CPT-sensitive difference.
chi-squared	109194.852	dof = 300
p-value	0.0000e+00	*** (p < 0.001)
lambda min	2.505e-12 m	
lambda max	1.629e-05 m	

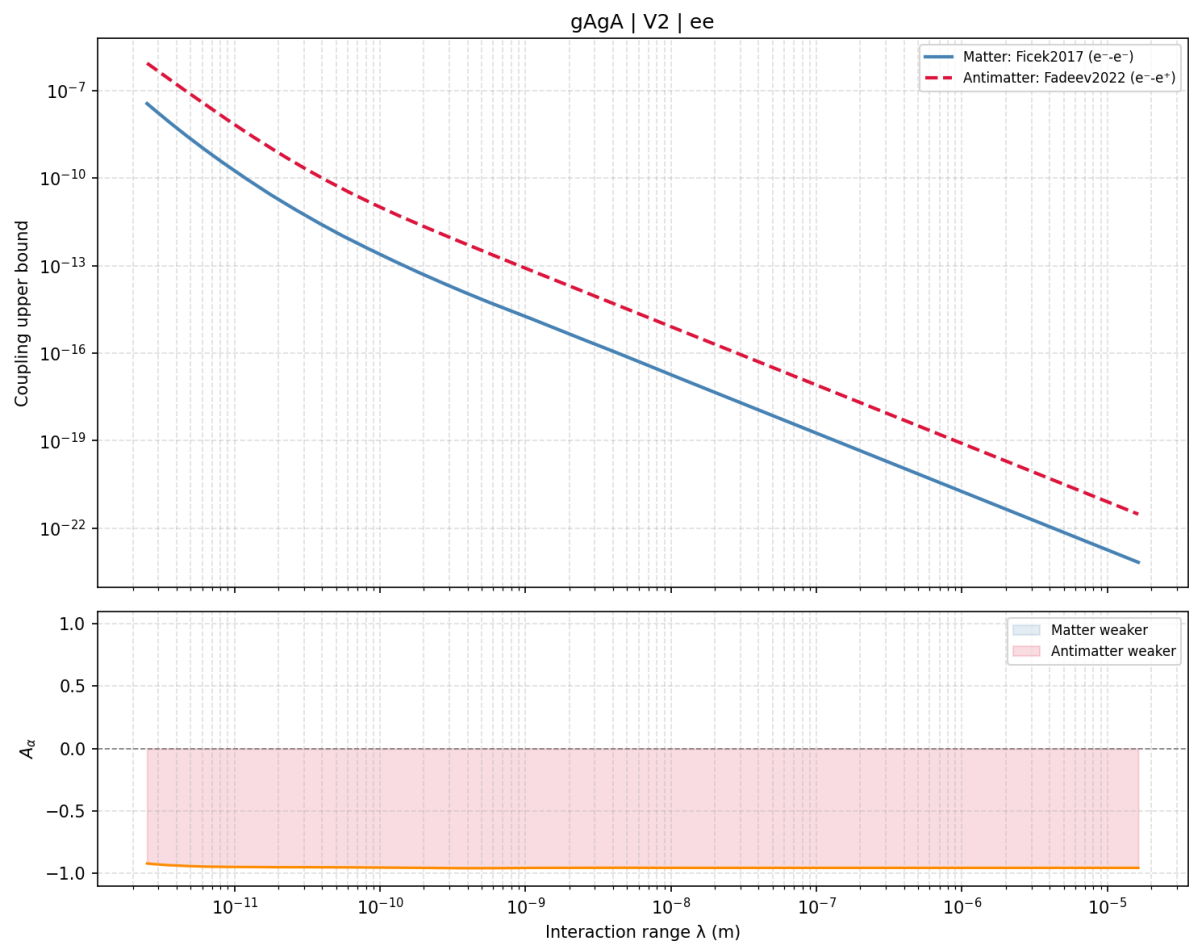


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under V2 potential.

Pair 6/7 | gAgA · V2 · ee

Matter dataset	Rong2018	Antimatter dataset	Fadeev2022
Interaction class	lepton-lepton	Lambda range	4.06e-06 – 1.63e-05 m

Statistic	Value	Interpretation
Mean A_alpha	1.0000	Strong asymmetry — large CPT-sensitive difference.
chi-squared	119992.939	dof = 300
p-value	0.0000e+00	*** (p < 0.001)
lambda min	4.063e-06 m	
lambda max	1.629e-05 m	

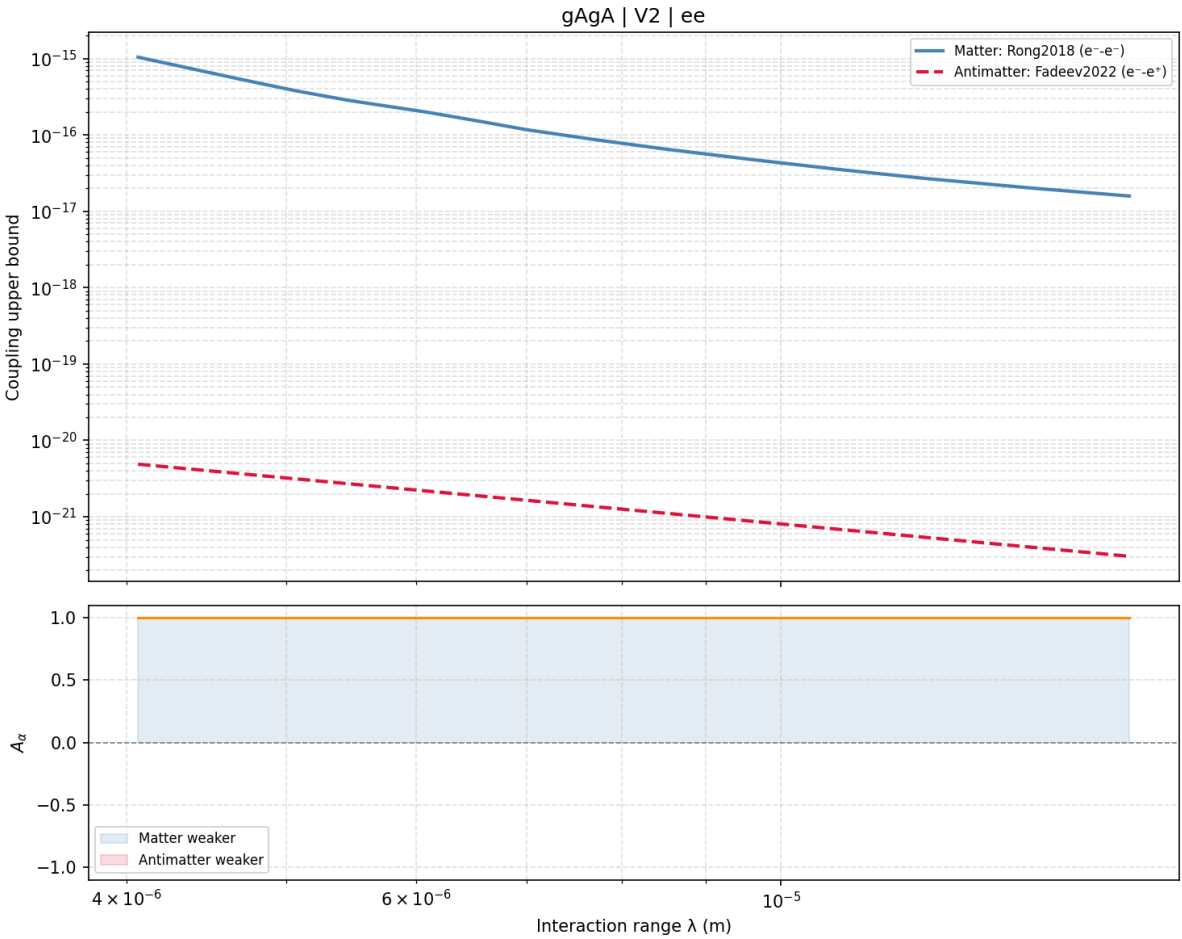


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under V2 potential.

Pair 7/7 | gAgA · V2 · ee

Matter dataset	Jiao2019	Antimatter dataset	Fadeev2022
Interaction class	lepton-lepton	Lambda range	3.24e-09 – 3.83e-07 m

Statistic	Value	Interpretation
Mean A_alpha	1.0000	Strong asymmetry — large CPT-sensitive difference.
chi-squared	119989.863	dof = 300
p-value	0.0000e+00	*** (p < 0.001)
lambda min	3.242e-09 m	
lambda max	3.825e-07 m	

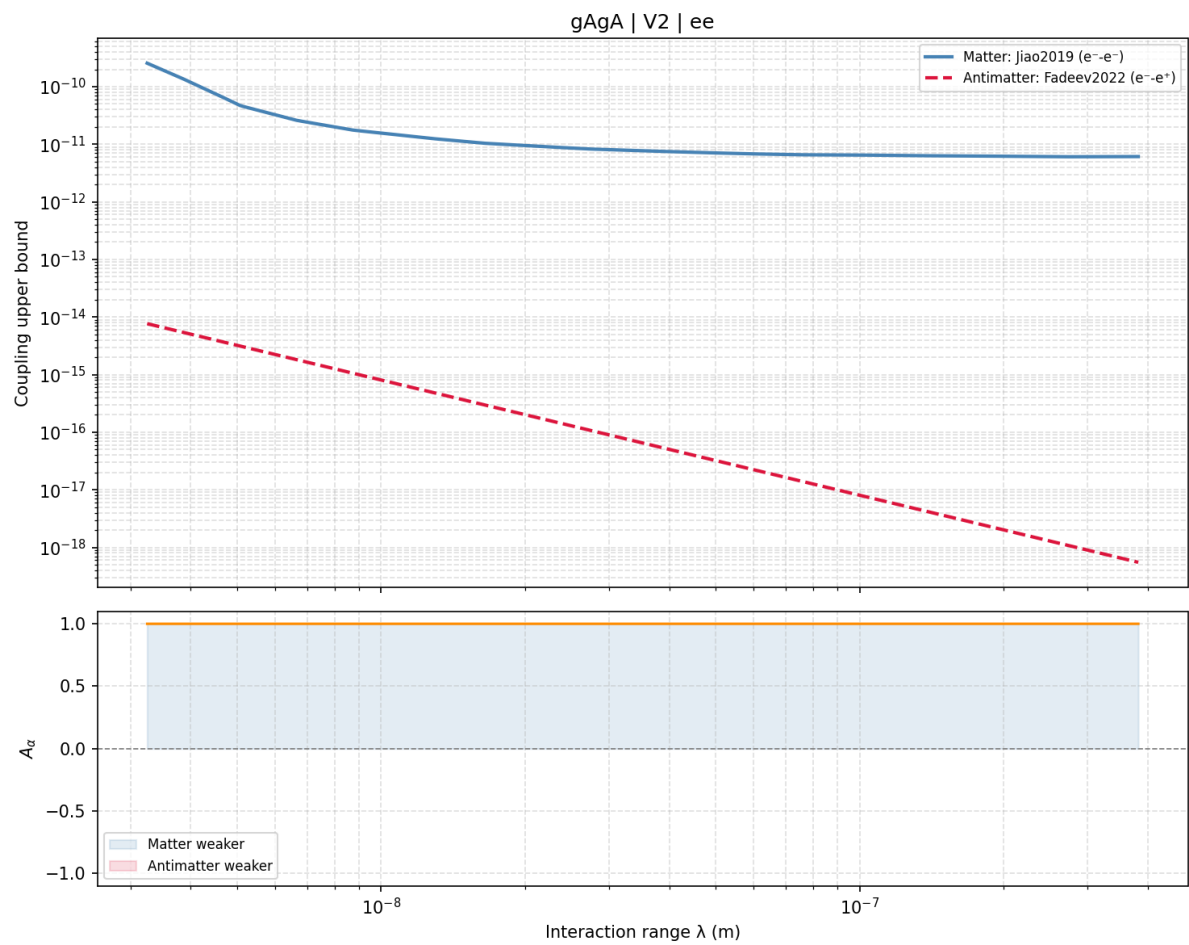


Figure: Coupling upper bounds vs interaction range λ (top panel) and asymmetry parameter A_α (bottom panel) for the ee sector under V2 potential.